**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 098 · Secondary Storage

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why might an operating system use virtual memory?

Q2. Which is a benefit of a high-level language?
A. Never needs translation
B. Easier for humans to read
C. Runs directly on every CPU
D. Uses no programming rules

Q3. Convert binary 10010110 to hexadecimal, using four-bit groups.

Q4. Use decimal units. A file is 4 MB. Give its size in KB and bytes.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Magnetic storage — Stores bits as magnetic patterns.

Optical storage — Uses lasers to read data from discs.

Solid-state storage — Non-volatile storage with no moving parts.

Cost per GB — Price per unit of stored capacity.

## New learning

Magnetic: An HDD stores data magnetically on rotating platters; it has moving mechanical parts.

Optical: Discs such as DVDs are read by lasers interacting with a reflective surface.

Solid state: SSDs and USB flash drives use non-volatile flash memory with no moving parts.

Choose for context: Compare capacity, cost per GB, speed, portability, reliability and durability; justify trade-offs.

## Worked example

A photographer carries equipment between locations. Explain one advantage and one disadvantage of choosing an SSD instead of an HDD. [4 marks]

1. Identify the use case

2. The drive will travel often and hold large files.

3. SSD has no moving parts and is more resistant to impacts; access can also be faster.

4. An SSD may cost more per GB than an HDD; justify one trade-off. [4]

# We Do · Practise with guidance

### Guided 1

Choose storage for a large, low-cost archive that is rarely moved. Explain your choice. [4]

Worked method: Magnetic HDD is suitable [1] for high capacities [1] at relatively low cost per GB [1]; slower access may be acceptable for infrequent use [1].

### Guided 2

A school sends files on DVDs. Identify the technology and explain a limitation. [3]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: Optical storage [1]; the disc is read using a laser [1] and may have limited capacity compared with modern SSDs or HDDs [1].

**Guided 3**

Explain why an SSD might be suitable for a moving laptop. [3]

Worked method: No moving parts [1] improves resistance to shocks [1], and fast read/write times [1] suit regular use.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

Calculate how many complete 400 MB files fit on a 2,000 MB drive.

```
capacityMB = 2000
fileMB = 400
files = ...
print(files)
```

write

## Question 1

Name the storage technology used by (i) DVD, (ii) HDD and (iii) USB flash drive. [3] Source: Original OCR J277-style practice question.

## Question 2

Explain one advantage and one disadvantage of magnetic storage compared with solid-state storage. [4] Source: Original OCR J277-style practice question.

## Question 3

A school needs portable, reusable storage for transporting files. Justify choosing a USB flash drive. [3] Source: Original OCR J277-style practice question.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Exit Ticket · Answer before checking

An algorithm only gives reliable results when its inputs are handled as specified.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. When there is not enough free RAM, secondary storage is used as virtual memory.

Q2. B — Easier for humans to read

Q3. 96: 1001 0110.

Q4. (a) 4,000 KB and 4,000,000 bytes.

## I Do / We Do pages · Guided practice

Q1. Magnetic HDD is suitable [1] for high capacities [1] at relatively low cost per GB [1]; slower access may be acceptable for infrequent use [1].

Q2. Optical storage [1]; the disc is read using a laser [1] and may have limited capacity compared with modern SSDs or HDDs [1].

Q3. No moving parts [1] improves resistance to shocks [1], and fast read/write times [1] suit regular use.

## You Do pages · Code First

```
capacityMB = 2000
fileMB = 400
files = capacityMB DIV fileMB
print(files)
```

Expected result: 5

## You Do pages · Independent questions

Q1. (i) Optical [1]; (ii) magnetic [1]; (iii) solid-state/flash [1].

Q2. Magnetic HDD often costs less per GB [1], advantageous for large capacity [1]; it has moving parts [1], increasing vulnerability to physical shock [1].

Q3. Solid-state [1], compact and portable [1], reusable with no moving parts [1].

## Exit Ticket · Model answer

True

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**